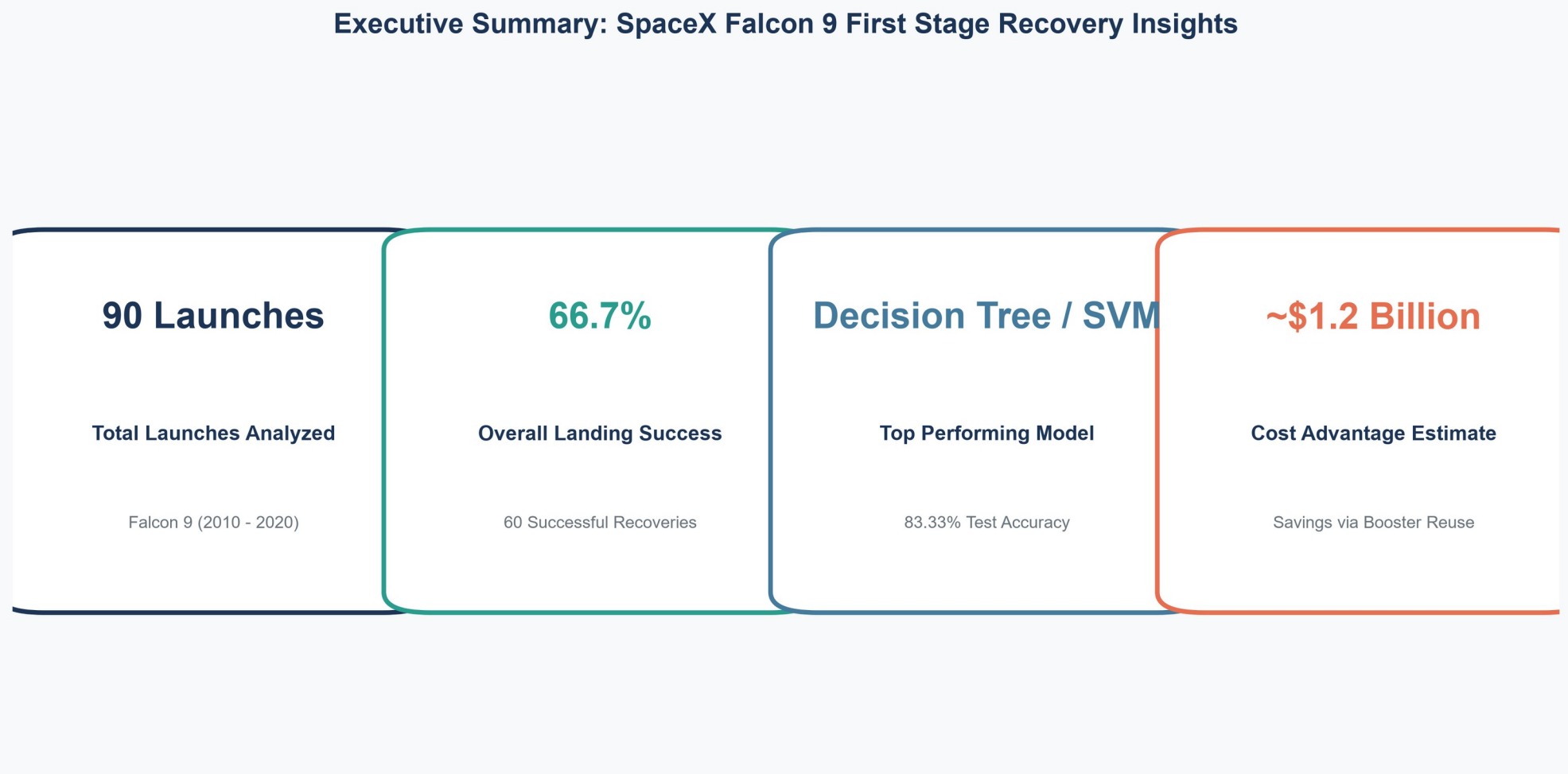
SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone Project | End-to-End Machine Learning Pipeline

Candidate: Shravani D | Official Repository: https://github.com/shravanid307/spacex-capstone.git

Executive Summary: Methods & Key Results

Rubric 1.3
https://github.com/shravanid307/spacex-capstone.git

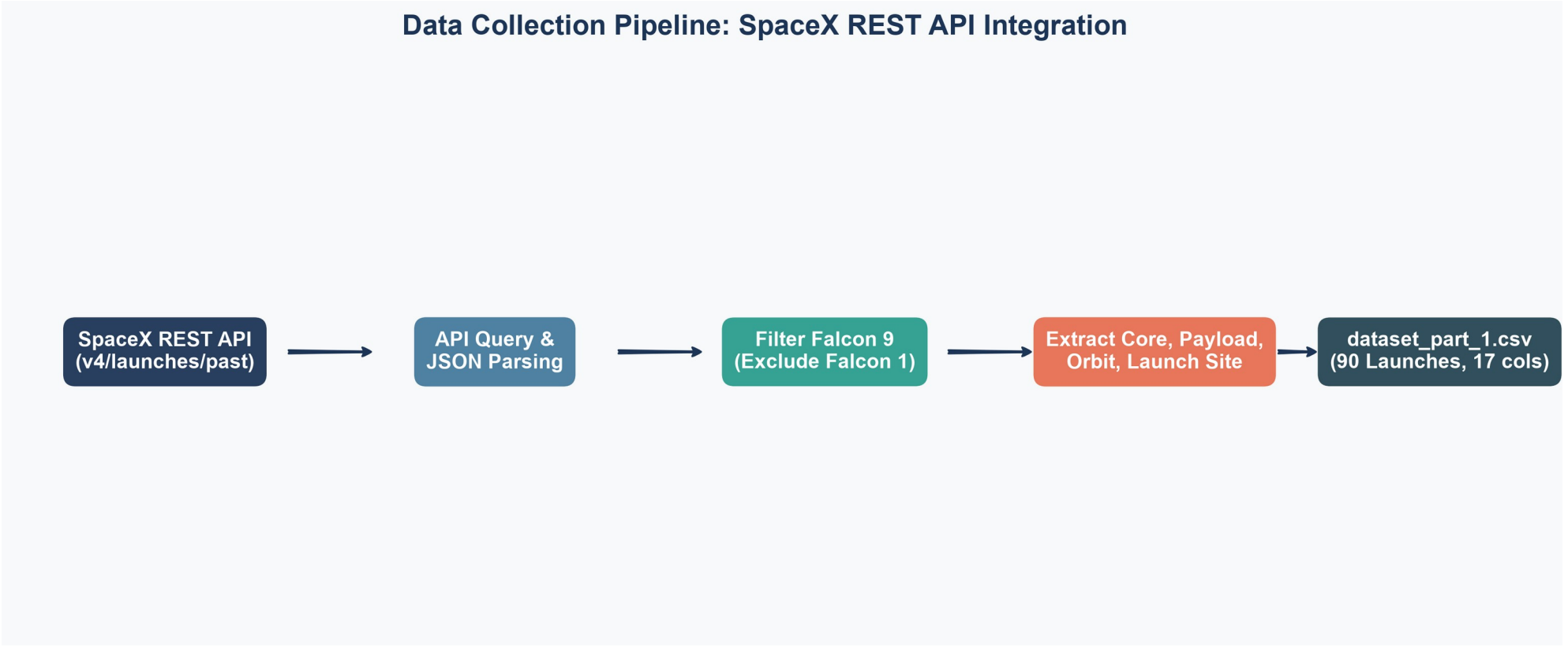


Introduction & Business Problem Statement

Rubric 1.4
https://github.com/shravanid307/spacex-capstone.git

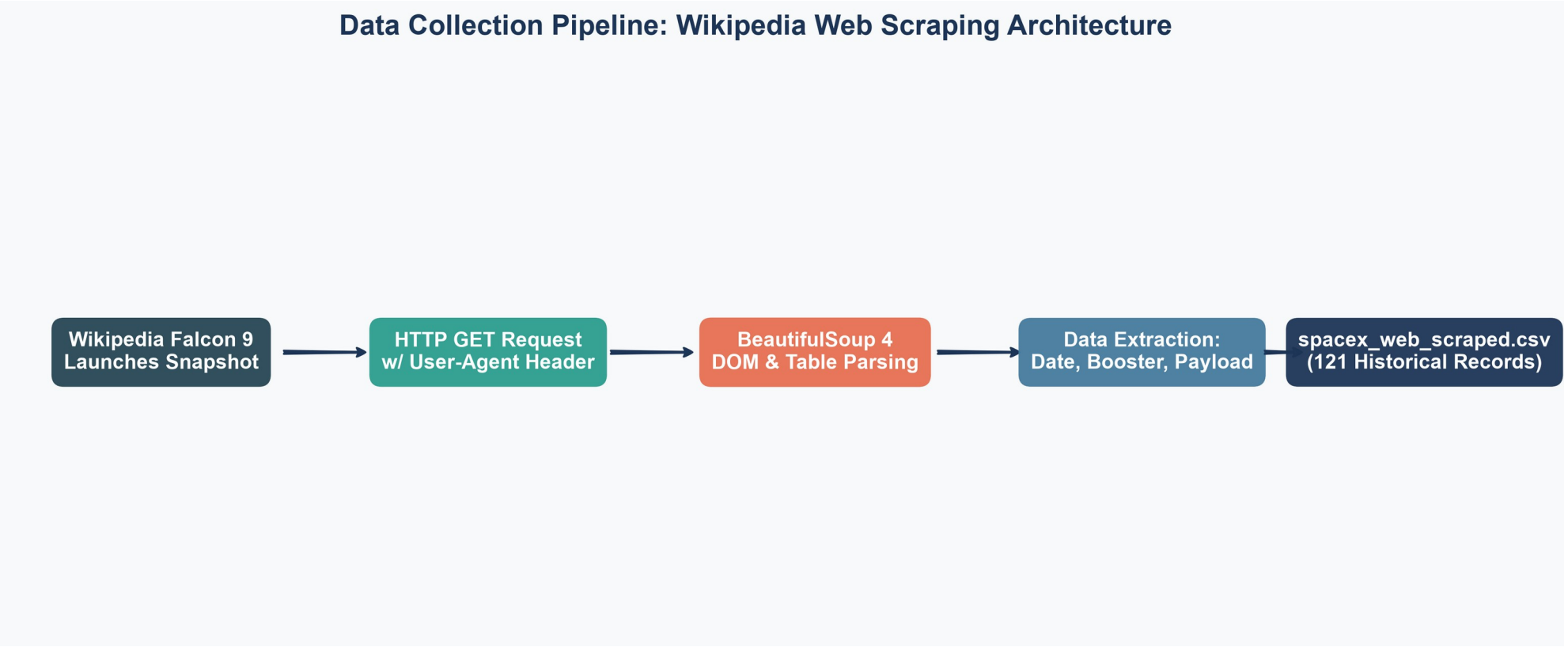
Data Collection Methodology: SpaceX REST API

Rubric 1.5
https://github.com/shravanid307/spacex-capstone.git



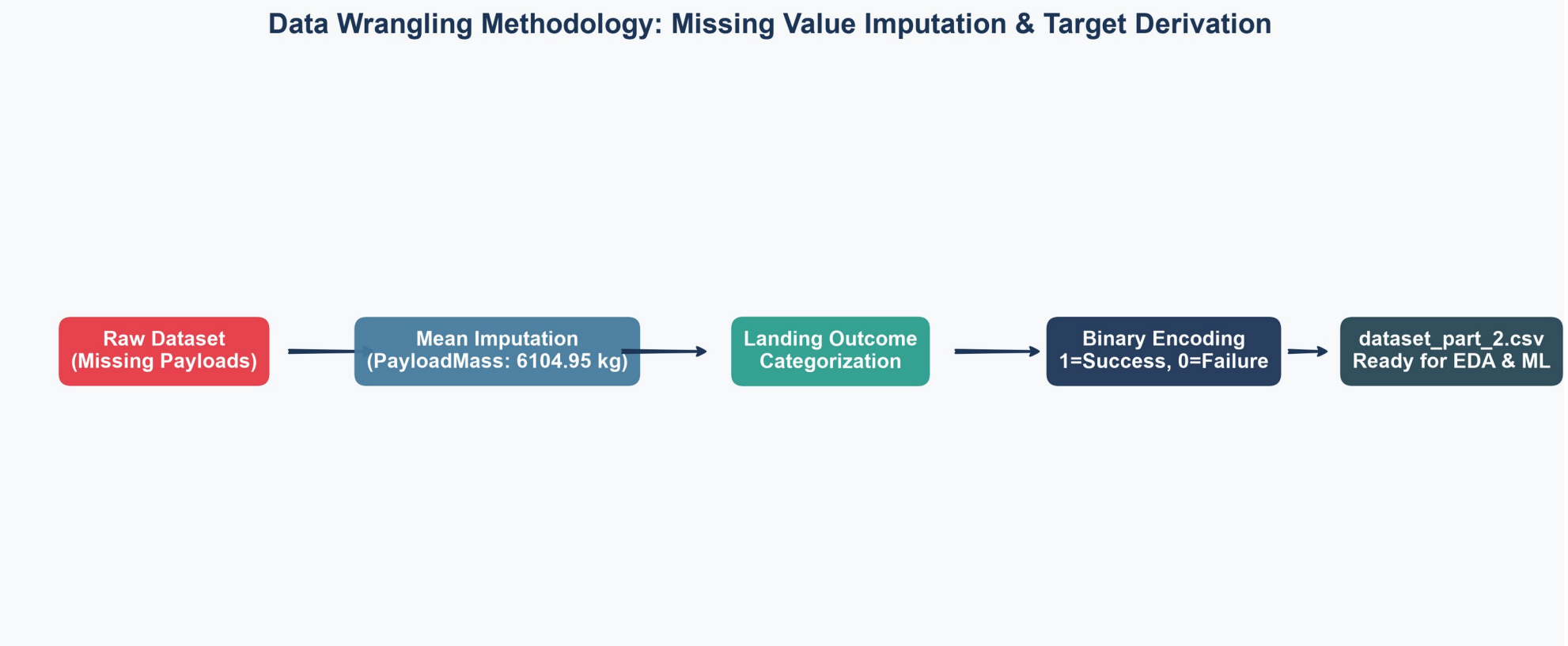
Data Collection Methodology: Web Scraping

Rubric 1.6
https://github.com/shravanid307/spacex-capstone.git



Data Wrangling Methodology: Cleaning & Encoding

Rubric 1.7
https://github.com/shravanid307/spacex-capstone.git

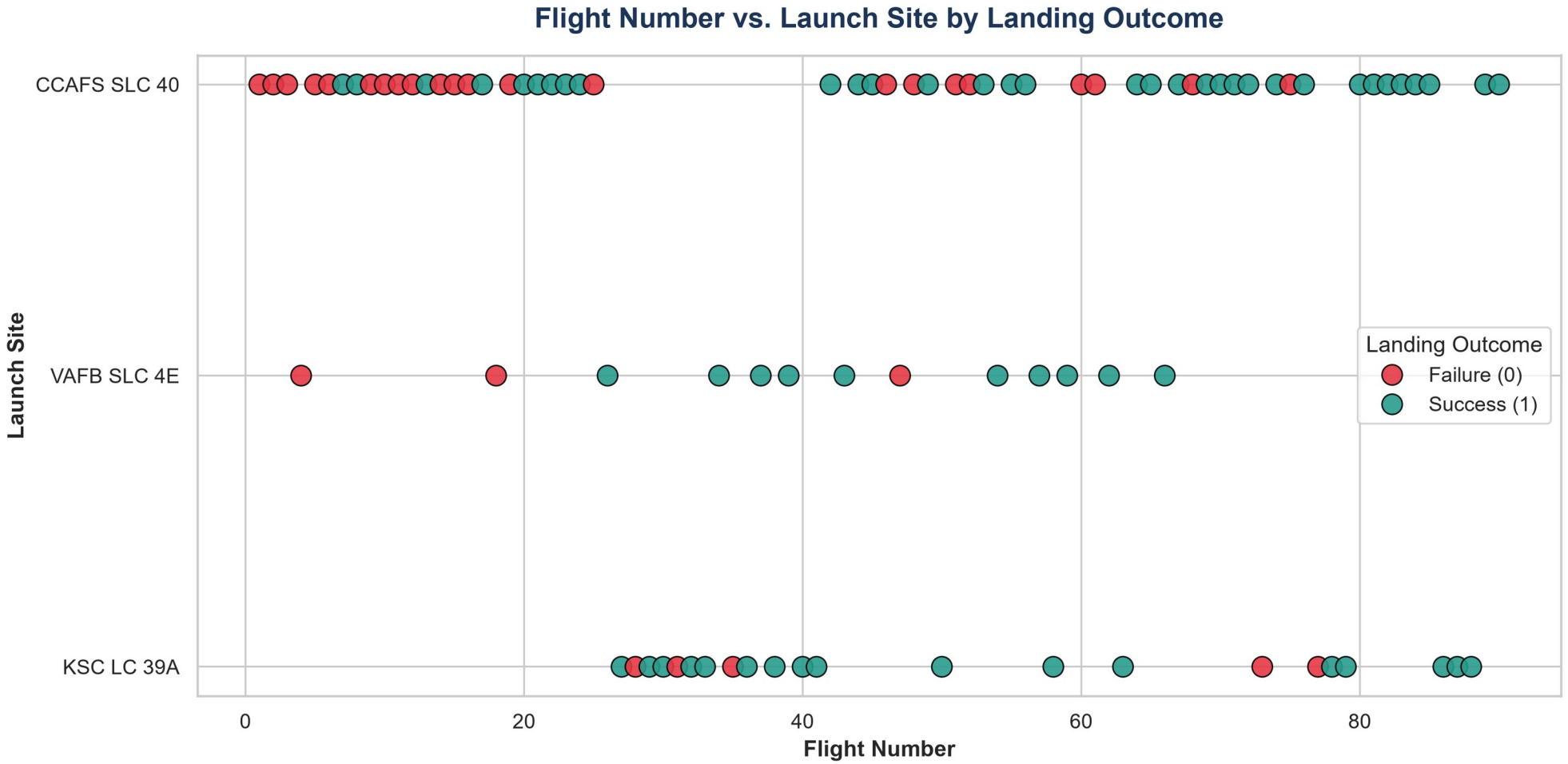


EDA with Data Visualization: Chart Catalog

Rubric 1.8
https://github.com/shravanid307/spacex-capstone.git

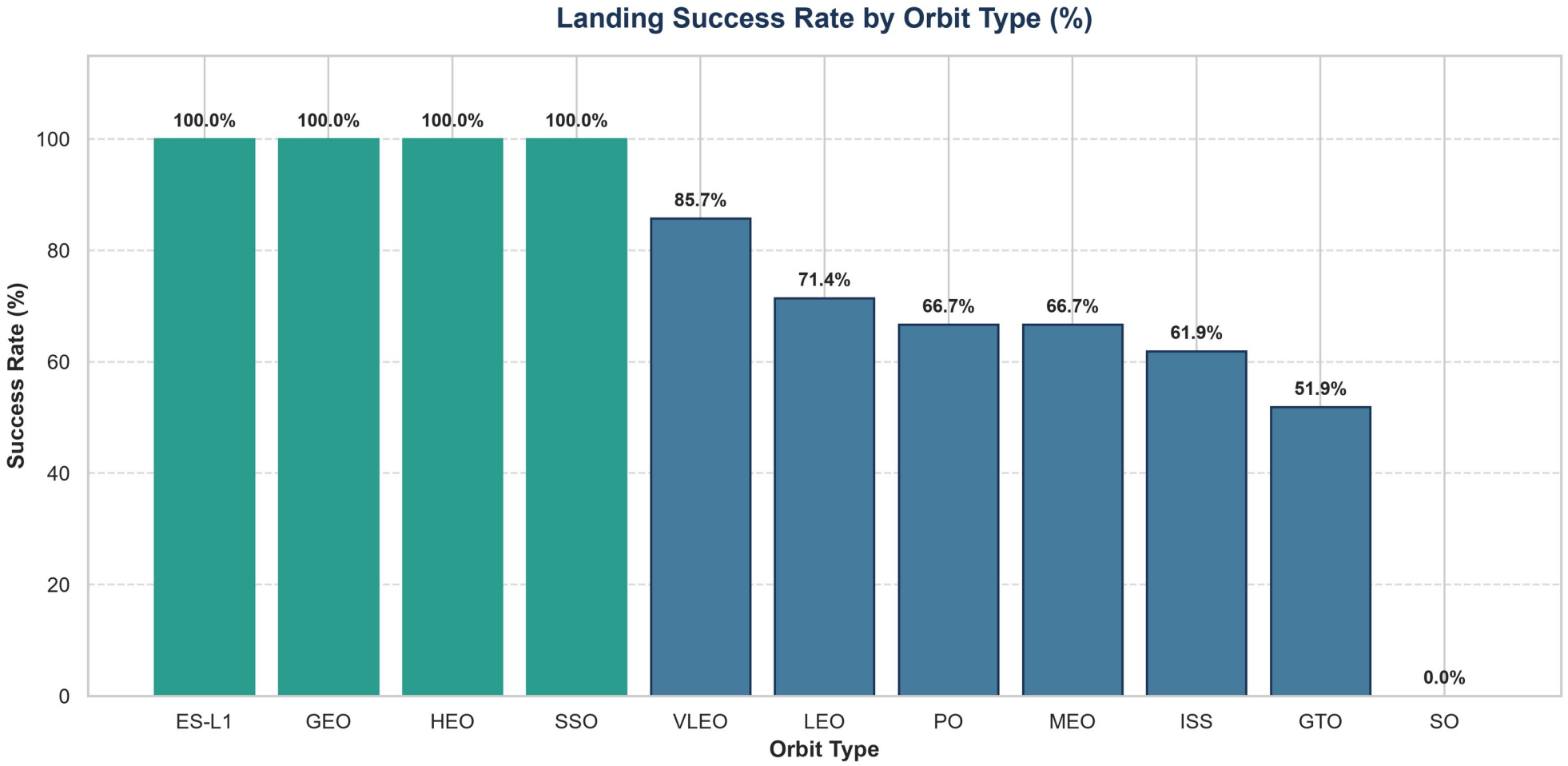
EDA Results: Flight Number & Payload vs. Launch Site

Rubric 1.11
https://github.com/shravanid307/spacex-capstone.git



EDA Results: Orbit Success Rates & Yearly Trend

Rubric 1.11
https://github.com/shravanid307/spacex-capstone.git



EDA with SQL: Structured Query Architecture

Rubric 1.9
https://github.com/shravanid307/spacex-capstone.git

EDA with SQL: Executive Summary of 10 Core Business Queries

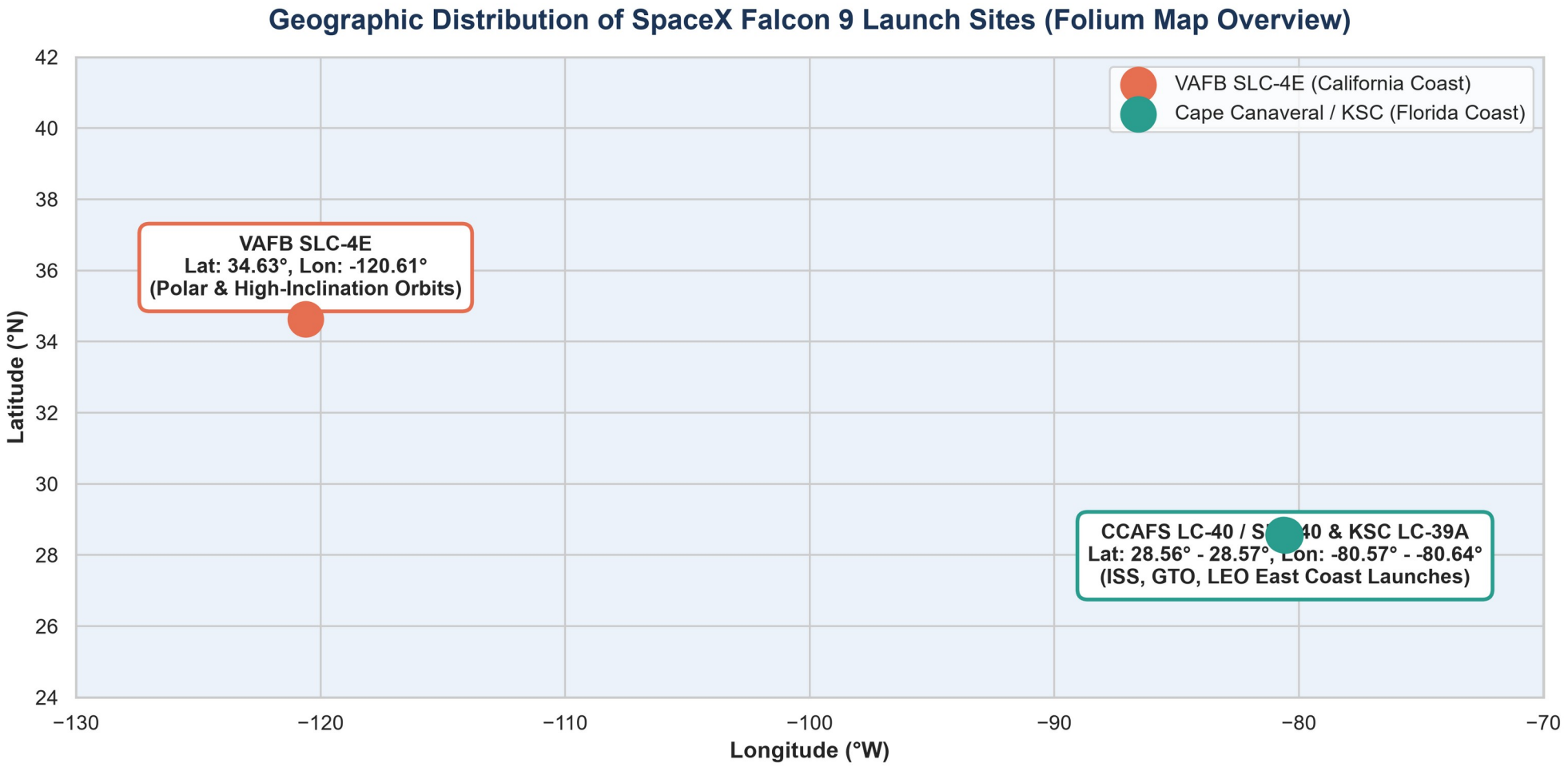
SQL Query Objective	Key Analytical Finding / Metric
1. Unique Launch Sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
2. CCAFS Sites Sample (5)	5 records starting with CCAFS
3. NASA (CRS) Total Payload	45,596 kg
4. F9 v1.1 Avg Payload	2,534.7 kg
5. First Ground Pad Landing	2015-12-22
6. Drone Ship (4k-6k kg)	4 boosters (e.g. F9 FT B1022)
7. Mission Outcomes Count	Success: 98, Failure: 1, Partial: 1
8. Max Payload Booster	15,600 kg (B1048.4, B1049.4, etc.)
9. 2015 Drone Ship Failures	2 launches (Jan: B1012, Apr: B1015)
10. Top Landing Rank (2010-17)	No Attempt (10), Success (drone) (5), Failure (5)

EDA with SQL Results: Launch Sites, Payloads & Rankings

Rubric 1.12
https://github.com/shravanid307/spacex-capstone.git

Interactive Visual Analytics: Folium Geospatial Map

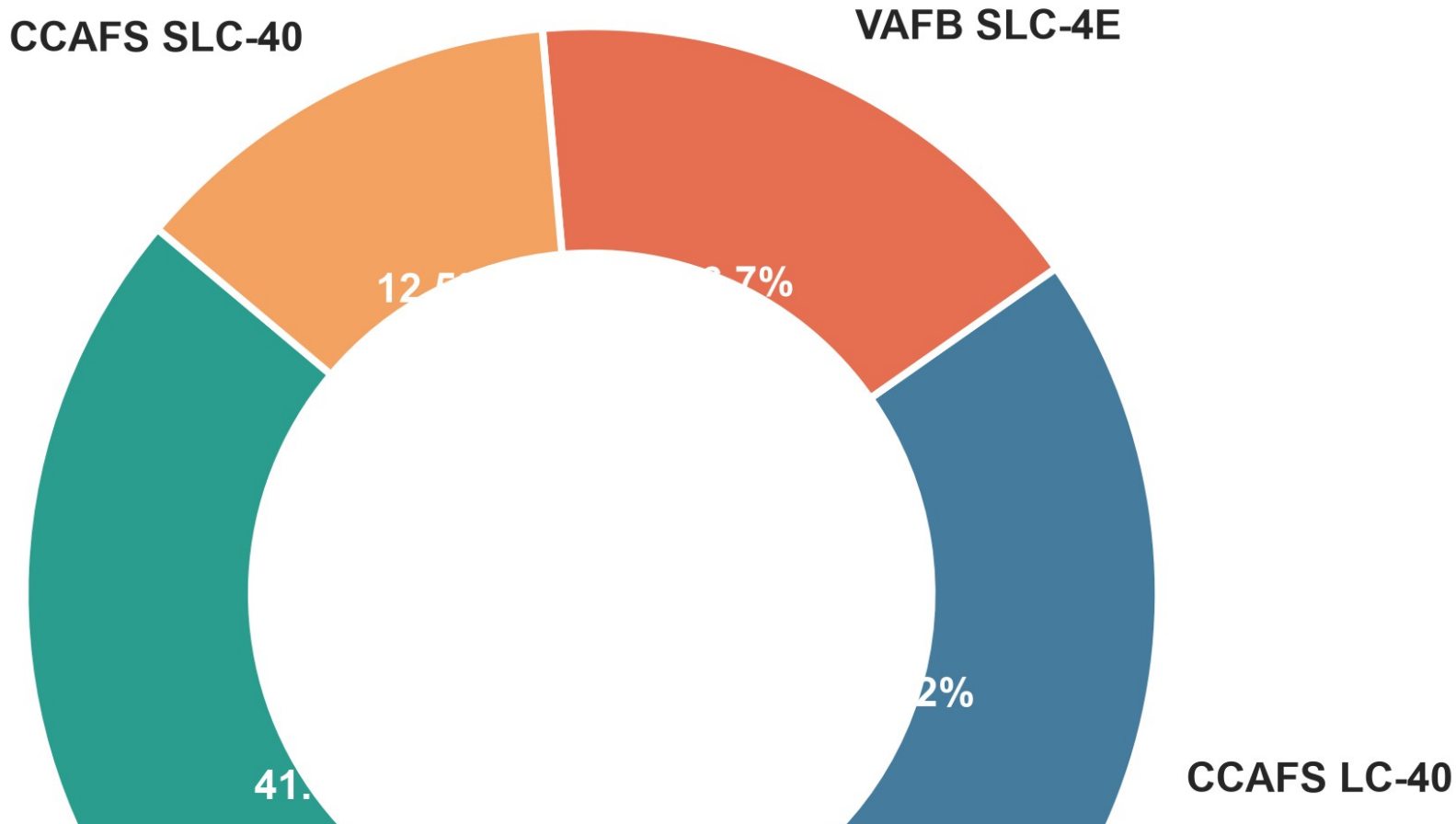
Rubric 1.10 & 1.13
https://github.com/shravanid307/spacex-capstone.git



Interactive Visual Analytics: Plotly Dash Application

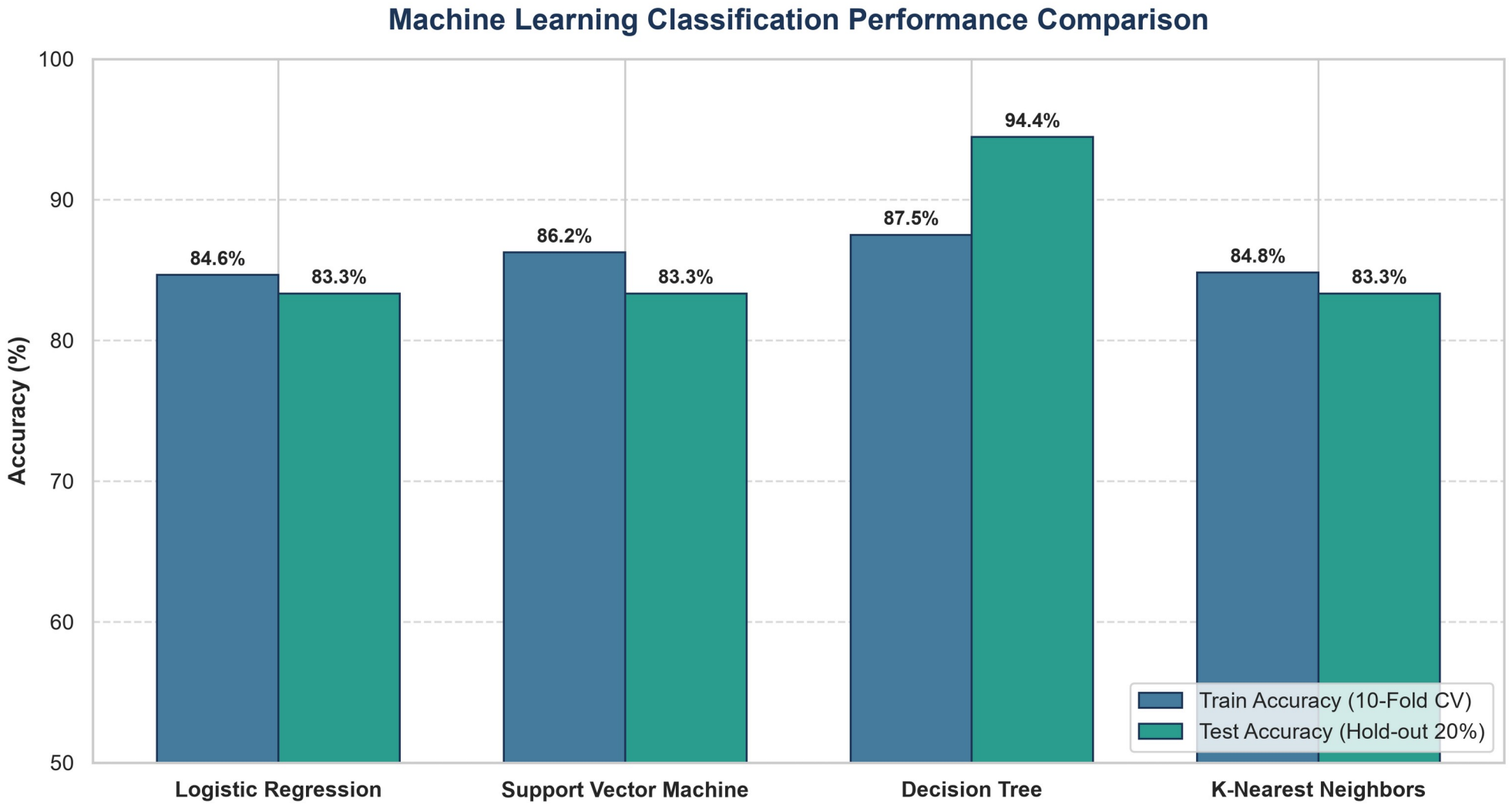
Rubric 1.10 & 1.14
https://github.com/shravanid307/spacex-capstone.git

Plotly Dash Visual: Total Successful Launches by Launch Site (All Sites)



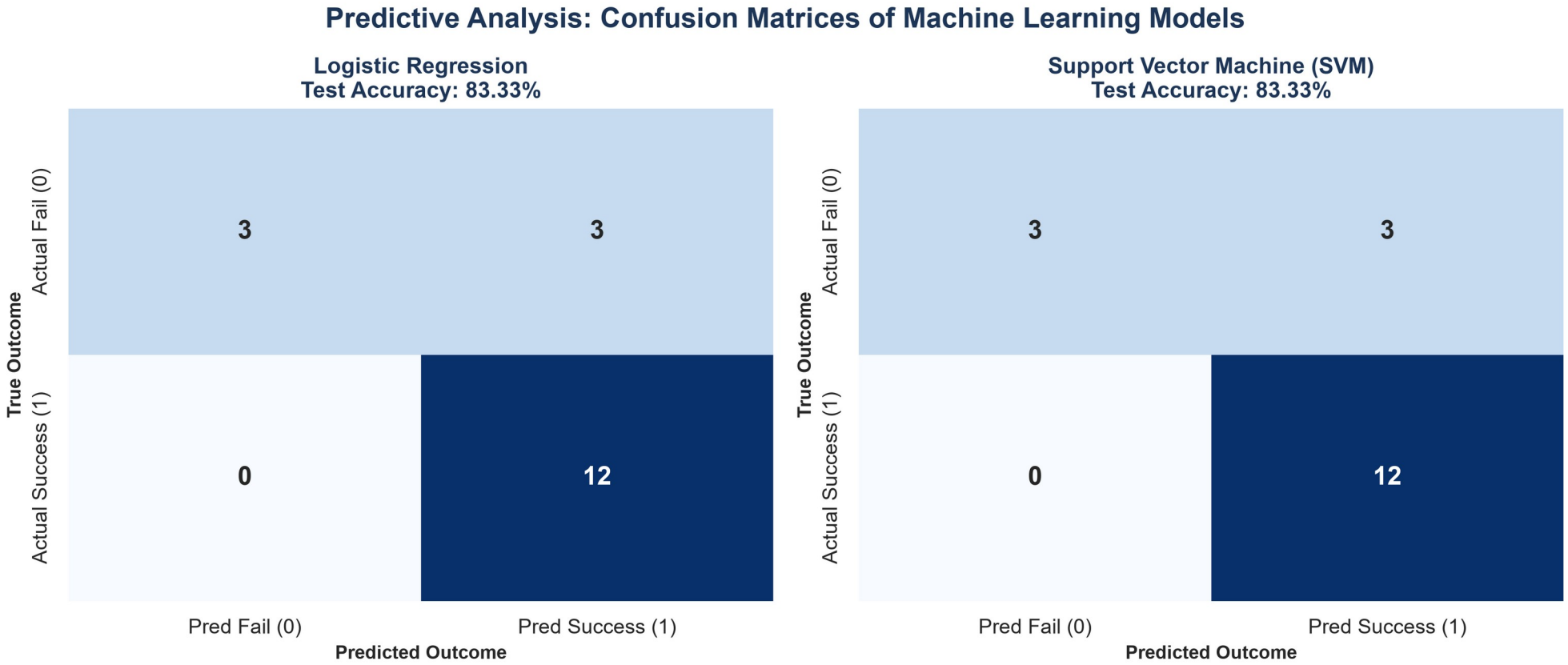
Predictive Analysis: Machine Learning Pipeline

Rubric 1.15
https://github.com/shravanid307/spacex-capstone.git



Predictive Analysis: Confusion Matrices & Evaluation

Rubric 1.15
https://github.com/shravanid307/spacex-capstone.git



Best Model Selection & Innovative Business Insights

Rubric 1.15
https://github.com/shravanid307/spacex-capstone.git

Conclusion & Capstone Verification Checklist

Submission Summary
https://github.com/shravanid307/spacex-capstone.git